Data Cleaning Report

Source: DOC-20261102-WA0001.csv | Retail order data, 20 columns, orders dated 2020-01-01 to 2024-12-30

1. Summary

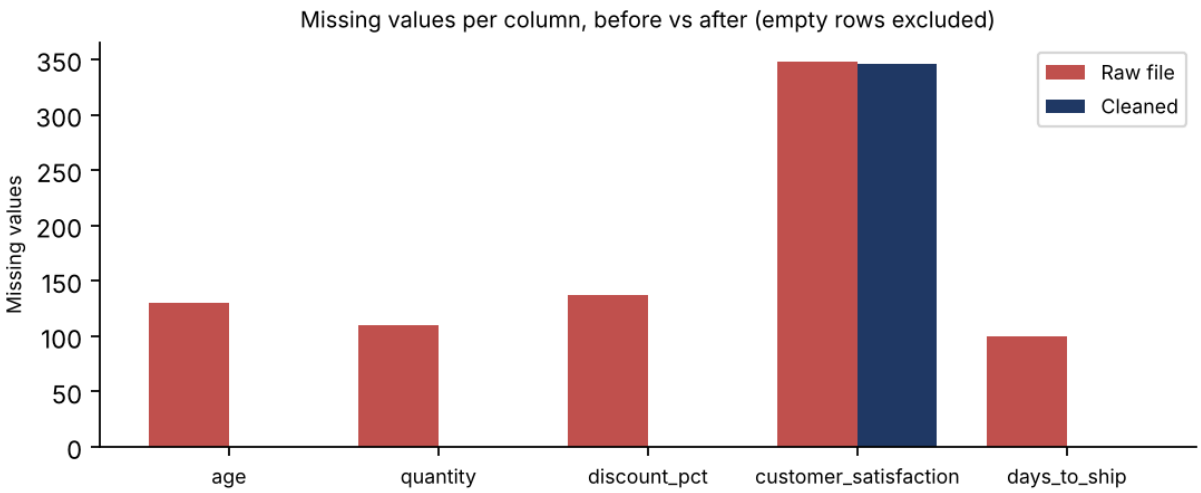
The raw file held **4,310** rows. After cleaning, **4,200** unique orders remain. **30** rows were completely empty and **80** were exact duplicates of another order. The remaining rows had mixed date formats, inconsistent text labels, impossible values (such as age 999 or quantity 999) and missing values. The most damaging problem was eight orders with a placeholder quantity of 999, which added about 31.6 million to total sales (the largest single order was 17.4 million against a realistic maximum of 1.18 million).

Measure	Raw file	Cleaned file
Rows	4,310	4,200
Unique order_id values	4,200 (plus 30 blank, 80 repeated)	4,200
Total sales amount (raw includes duplicates)	266,894,969	232,651,921
Largest single order	17,360,314	1,182,709
Columns with missing values	5 of 20 (age, quantity, discount_pct, customer_satisfaction, days_to_ship)	1 of 20 (customer_satisfaction, left blank by design)
Date format	3 formats	YYYY-MM-DD

2. Issues found and actions taken

#	Issue	Column	Rows	Action
1	Completely empty rows	(all)	30	Removed
2	Exact duplicate orders (same order_id, identical values)	order_id	80	Removed, first occurrence kept
3	Dates in DD/MM/YYYY format	order_date	147	Parsed as day-first (86 values had day >12, none had month >12) and converted to YYYY-MM-DD
4	Dates in "Month DD, YYYY" text format	order_date	120	Converted to YYYY-MM-DD
5	Inconsistent gender labels (male, MALE, f, M, ...)	gender	431	Standardised to Male / Female / Other
6	Inconsistent order_status casing	order_status	152	Standardised to title case
7	Impossible ages (negative or 999)	age	25	Set to missing, then imputed
8	Negative shipping cost	shipping_cost	2	Set to missing, then imputed
9	Impossible days_to_ship (negative or 100)	days_to_ship	9	Set to missing, then imputed
10	Invalid quantity (0, -1 or placeholder 999)	quantity	26	Replaced with the median quantity for that product
11	Missing quantity	quantity	108	Replaced with the median quantity for that product
12	sales_amount / profit inflated by quantity = 999 (up to 17.4M per order)	sales_amount, profit	8	Rescaled by (imputed quantity / 999)
13	Missing values	age	125	Filled with column median (34)
14	Missing values	discount_pct	135	Filled with column median (0.21)
15	Missing values	days_to_ship	97	Filled with column median (6)
16	Missing customer satisfaction ratings	customer_satisfaction	346	Left blank (a subjective rating should not be invented)

Rows affected are counted after empty rows and duplicates were removed; one row can carry several fixes.



3. Method and decisions

• **Duplicates.** Every duplicate matched its original in all columns, so keeping the first occurrence loses no information. The removed rows are kept on the Removed_Duplicates sheet.

• **Dates.** Slash dates were read as day/month/year: 86 had a first part above 12 and none had a second part above 12, which rules out month/day.

• **Placeholder values.** A quantity of 999, an age of 999 and a 100-day shipping time are data-entry sentinels, not real observations. Negative or zero quantities, negative ages and negative shipping values are impossible. All were treated as missing.

• **Imputation.** Missing or invalid quantity uses the median for that product. Age (34), discount (21%) and days to ship (6) use the column median. These are standard, simple choices; they keep every order usable but slightly narrow the spread of those columns.

• **Sales and profit on 999-unit rows.** Those eight rows had sales and profit computed from 999 units, so both were scaled by (imputed quantity / 999). This assumes the original calculation was proportional to quantity. The affected rows are marked in the cleaning_notes column.

• **Customer satisfaction.** 346 orders have no rating. A rating is an opinion, so it was left blank instead of invented.

• **Traceability.** The Cleaned_Data sheet has a cleaning_notes column naming every fix made to each of the 503 rows that needed a value-level fix.

4. Checks passed

• order_id is unique across all 4,200 rows.

• return_flag is True exactly when order_status is Returned.

• Every date parses and falls between 2020-01-01 and 2024-12-30.

• No missing values remain except customer_satisfaction.

• Workbook row count reconciles with 4,310 - 30 - 80 = 4,200 (formula check on the Summary sheet).

5. Things to be aware of

• 239 customers are aged under 16 (as young as 5). The values are plausible numbers, so they were kept, but they may be unreliable.

• 87 customer_id values appear with more than one customer name, and gender does not always match first names (for example, a male-coded name with Female). These could not be corrected without a master customer list, so they were left as supplied.

• Gender has a very large Other group (about 34%), which is unusual and worth confirming with the data owner.

• sales_amount does not equal quantity x unit_price x (1 - discount) exactly (it is often 10-40% higher, and sometimes lower), so it likely includes tax or other charges. It was not recalculated.